



铁头山羊STM32教程

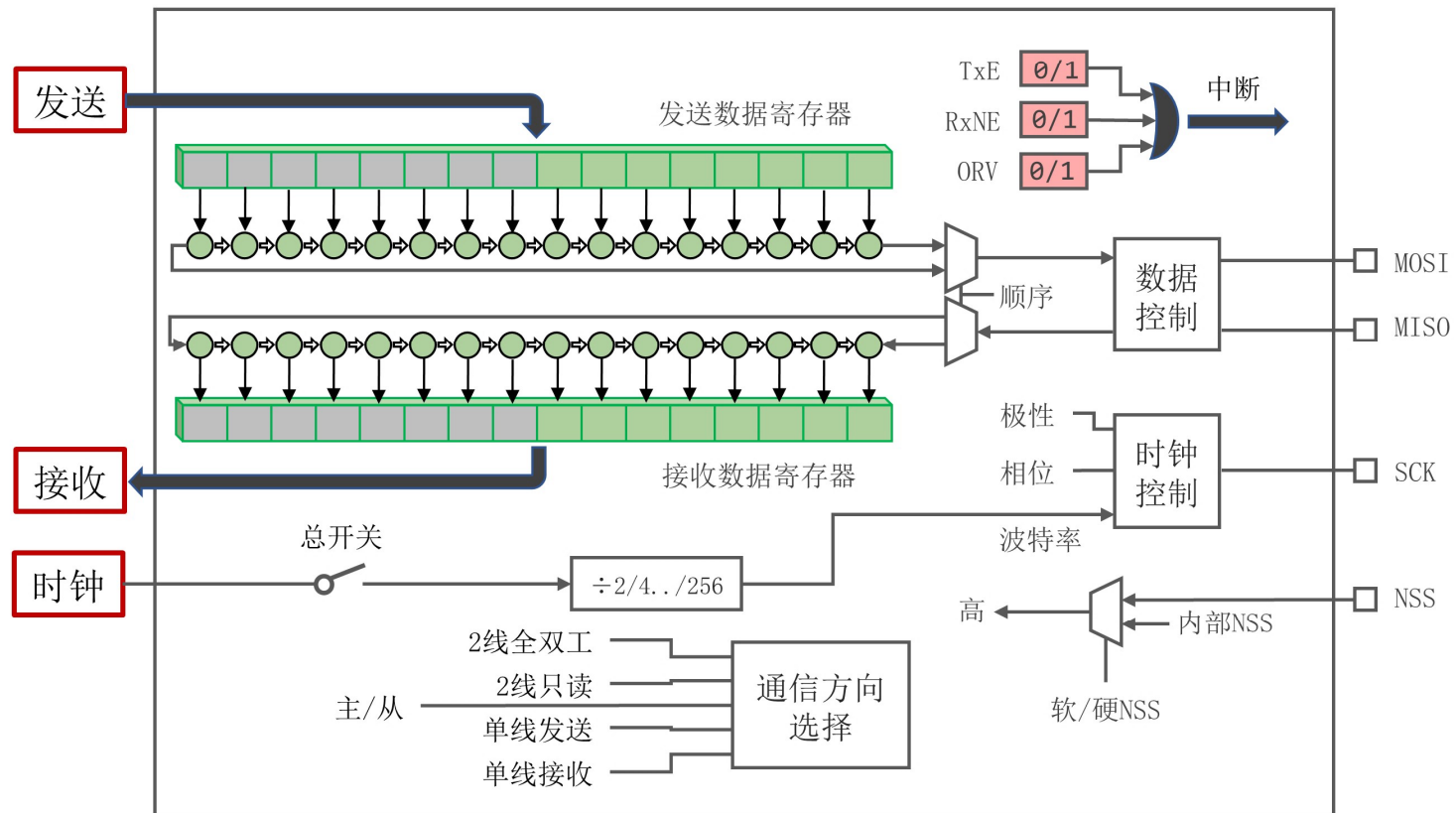
[SPI]

5.5. SPI 模块的初始化

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- 1. SPI模块的基本工作原理
- 2. SPI_Init
- 3. 选择数据通信方向
- 4. 数据宽度、极性、相位和比特位传输顺序
- 5. 设置波特率
- 6. NSS的配置方式



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```
void SPI_Init(SPI_TypeDef *SPIx, // SPI的名称, 可以是SPI1或SPI2  
              SPI_InitTypeDef *SPI_InitStruct); // 用于传递初始化参数
```

作用: 对SPI模块进行初始化

```
struct SPI_InitTypeDef{  
    uint16_t SPI_Direction; // 用来选择SPI通信的方向  
    uint16_t SPI_Mode;      // 用来选择SPI的模式, SPI_Master - 主机, SPI_Slave - 从机  
    uint16_t SPI_DataSize;  // 数据宽度, SPI_DataSize_8b - 8bit, SPI_DataSize_16b - 16bit  
    uint16_t SPI_CPOL;      // 时钟的极性  
    uint16_t SPI_CPHA;      // 时钟的相位  
    uint16_t SPI_NSS;       // 软件NSS/硬件NSS  
    uint16_t SPI_BaudRatePrescaler; // 用来选择波特率分频器的分频系数  
    uint16_t SPI_FirstBit;  // 比特位的传输顺序, SPI_FirstBit_MSB, SPI_FirstBit_LSB,  
}
```

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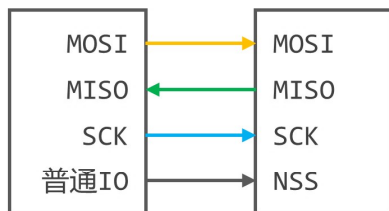
```
struct SPI_InitTypeDef{
```

```
...
```

```
uint16_t SPI_Direction; // 通信的方向 - SPI_Direction_2Lines_FullDuplex 2线全双工
//                               - SPI_Direction_2Lines_ReadOnly 2线只读
//                               - SPI_Direction_1Line_Rx 单线接收
//                               - SPI_Direction_1Line_Tx 单线发送
```

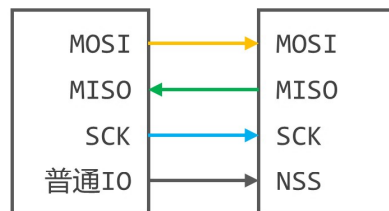
```
...
```

```
}
```



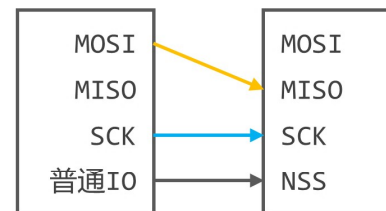
2线全双工

标准的SPI总线格式



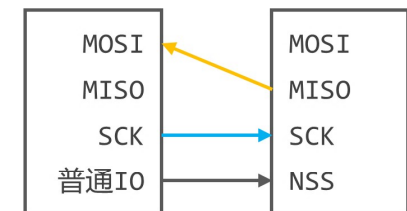
2线只读

一般用于从机



1线发送

单线，方向是发送



1线接收

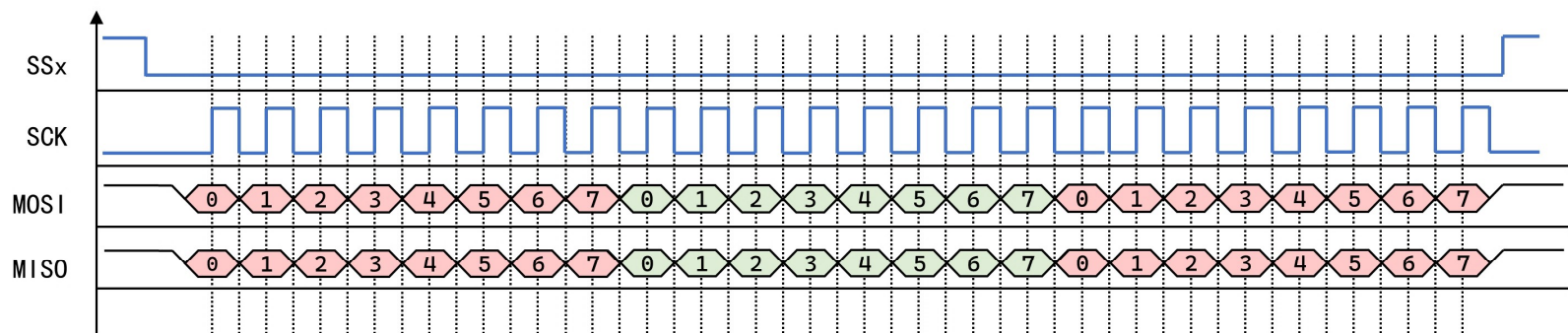
单线，方向是接收

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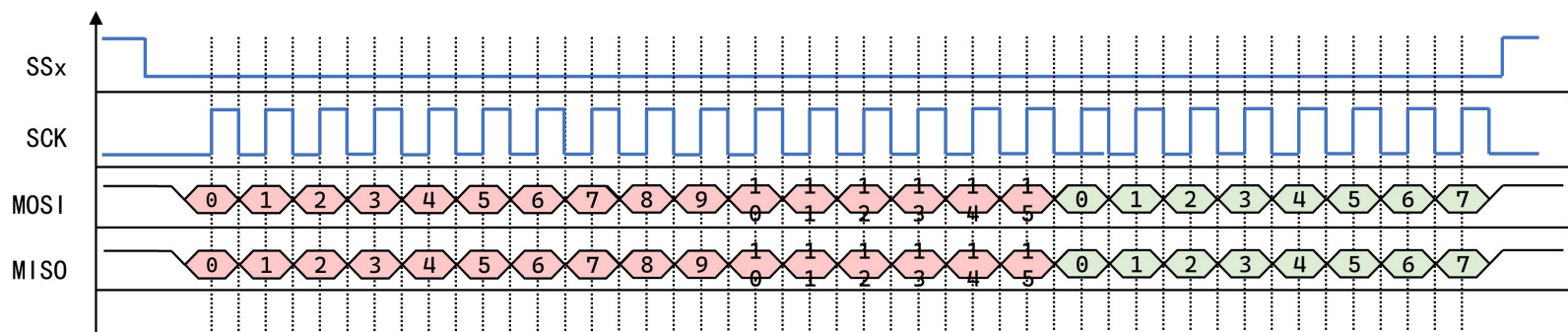
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数据宽度

8 bit



16 bit



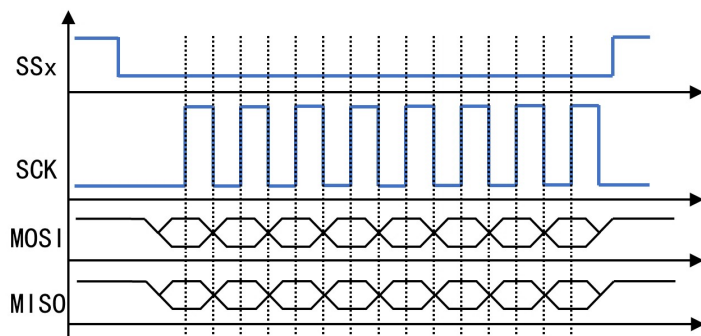
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极性和相位

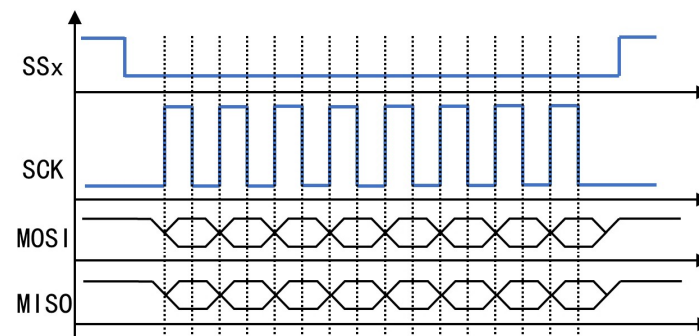
模式0

极性=0
相位=0



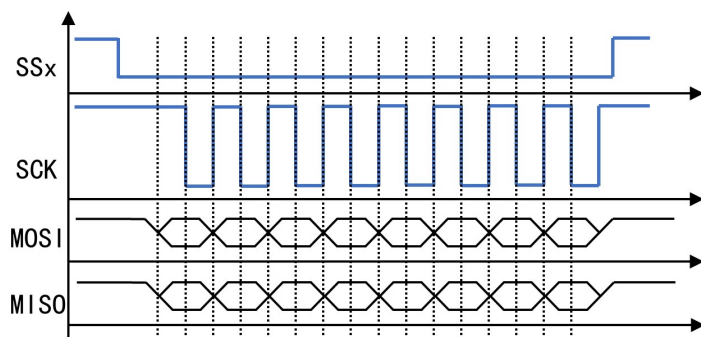
模式1

极性=0
相位=1



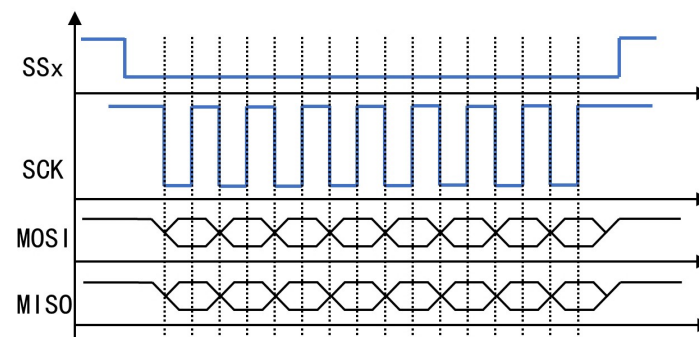
模式2

极性=1
相位=0



模式3

极性=1
相位=1

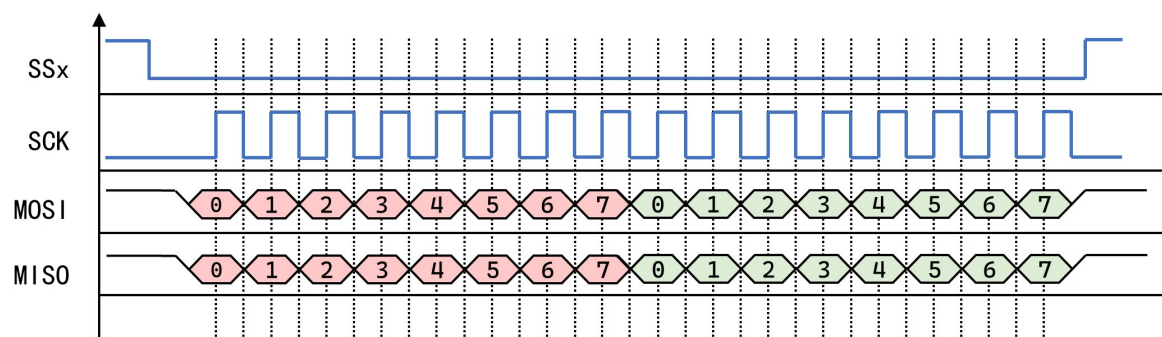


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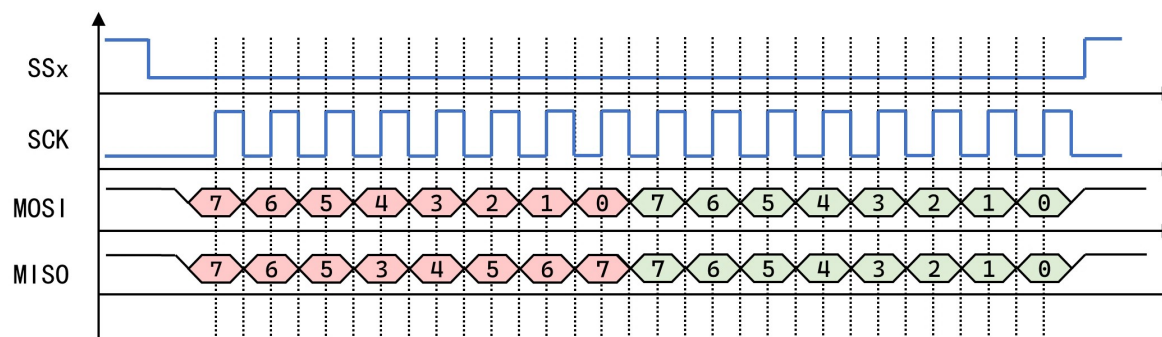
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比特位的传输顺序

LSB First
先传最低有效位

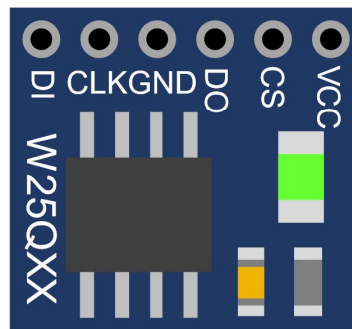


MSB First
先传最高有效位

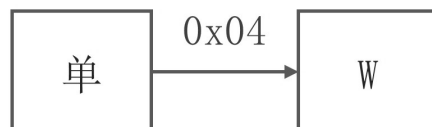


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11.2.5 Write Disable (04h)

The Write Disable instruction (Figure 5) resets the Write Enable Latch (WEL) bit in the Status Register to a 0. The Write Disable instruction is entered by driving /CS low, shifting the instruction code "04h" into the DI pin and then driving /CS high. Note that the WEL bit is automatically reset after Power-up and upon completion of the Write Status Register, Page Program, Sector Erase, Block Erase and Chip Erase instructions.

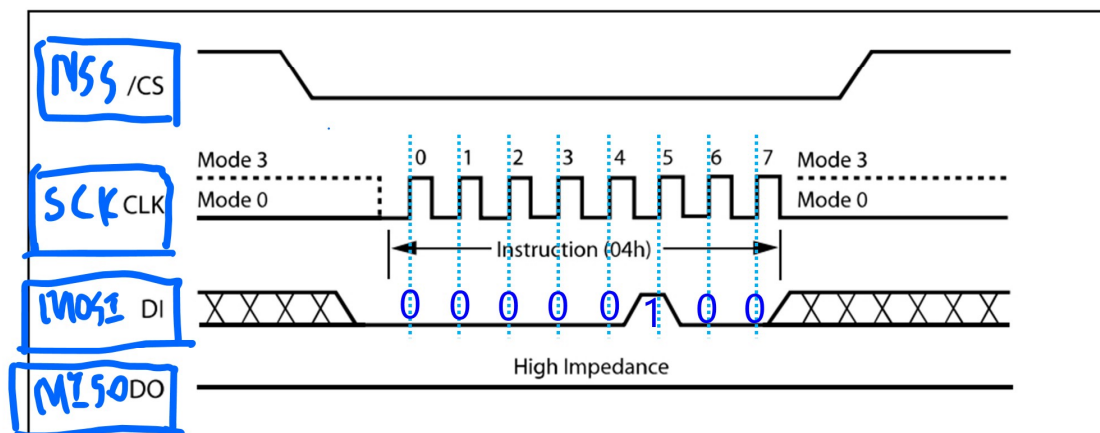


Figure 5. Write Disable Instruction Sequence Diagram

数据宽度

8b / 16b

极性

高 / 低

相位

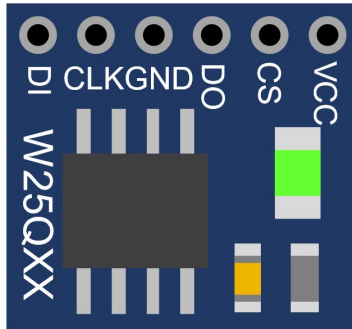
1 / 2

比特位传输顺序

MSB / LSB

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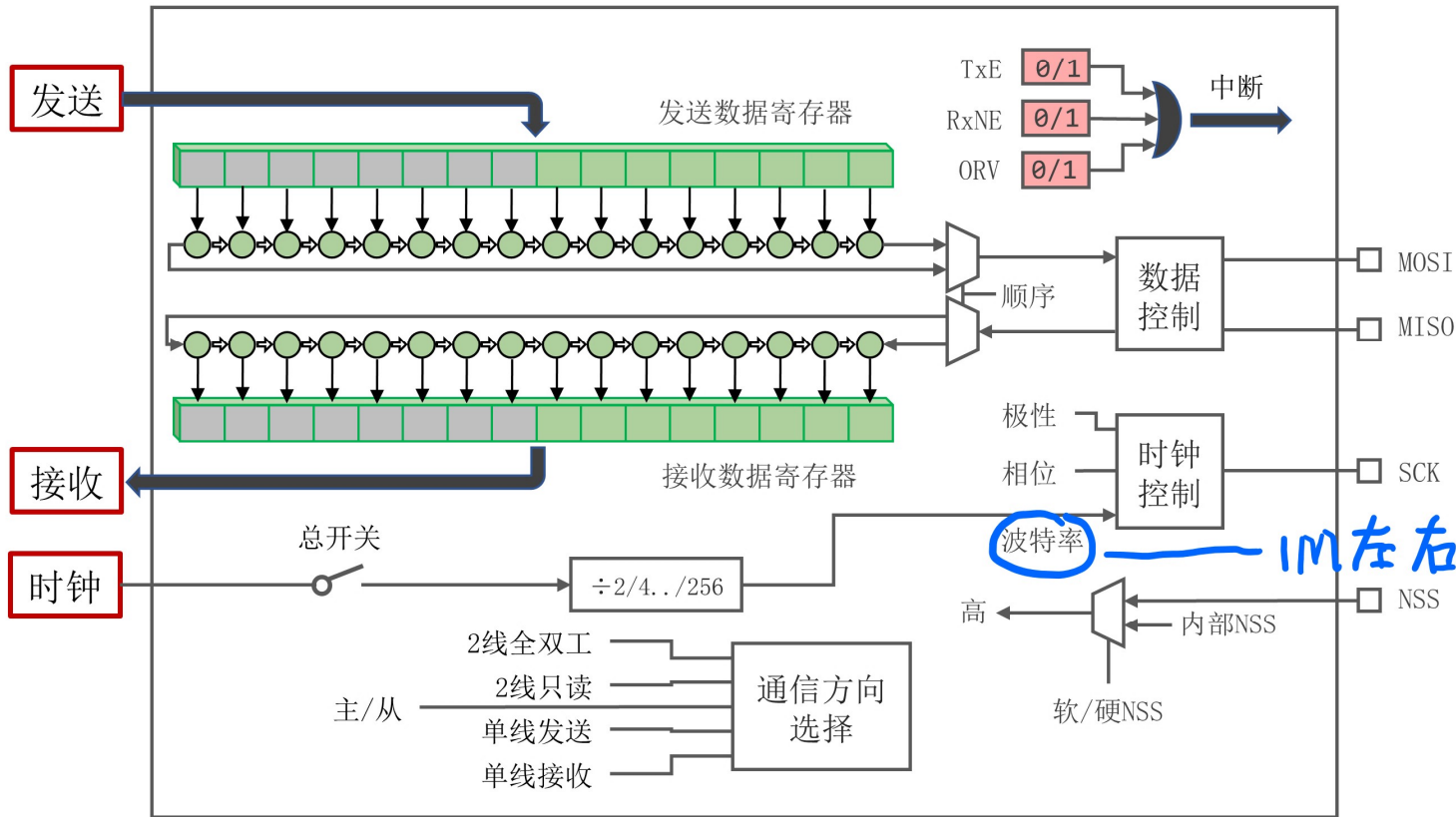
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- **Highest Performance Serial Flash**
 - Up to 6X that of ordinary Serial Flash
 - 80MHz clock operation
 - 160MHz equivalent Dual SPI
 - 320MHz equivalent Quad SPI
 - 40MB/S continuous data transfer rate

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$$72\text{M} \div 2 = 36\text{M}$$

$$72\text{M} \div 4 = 18\text{M}$$

$$72\text{M} \div 8 = 9\text{M}$$

$$72\text{M} \div 16 = 4.5\text{M}$$

$$72\text{M} \div 32 = 2.25\text{M}$$

$$72\text{M} \div 64 = 1.125\text{M}$$

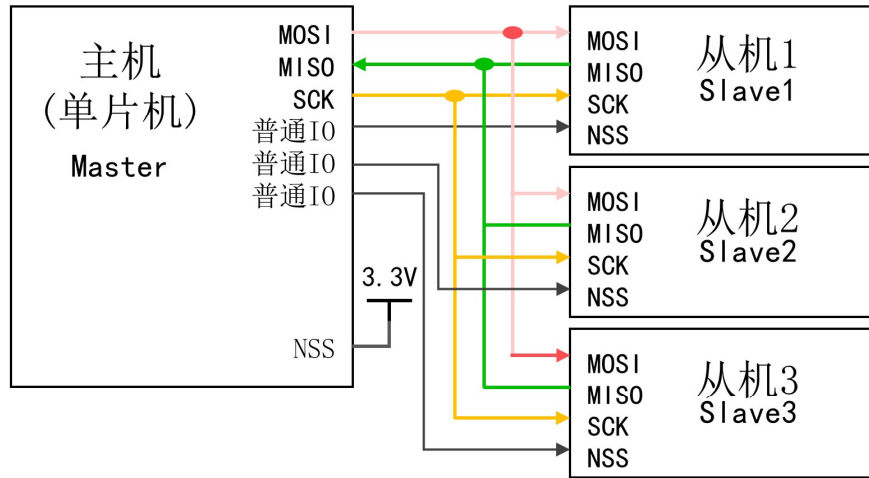
$$72\text{M} \div 128 = 0.5625\text{M}$$

$$72\text{M} \div 256 = 0.28125\text{M}$$

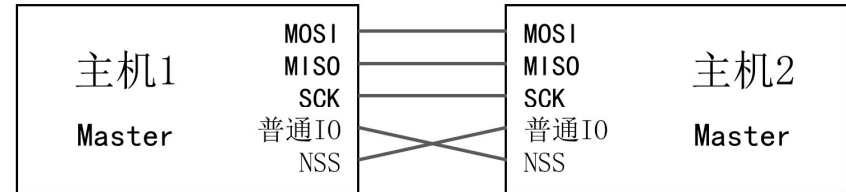
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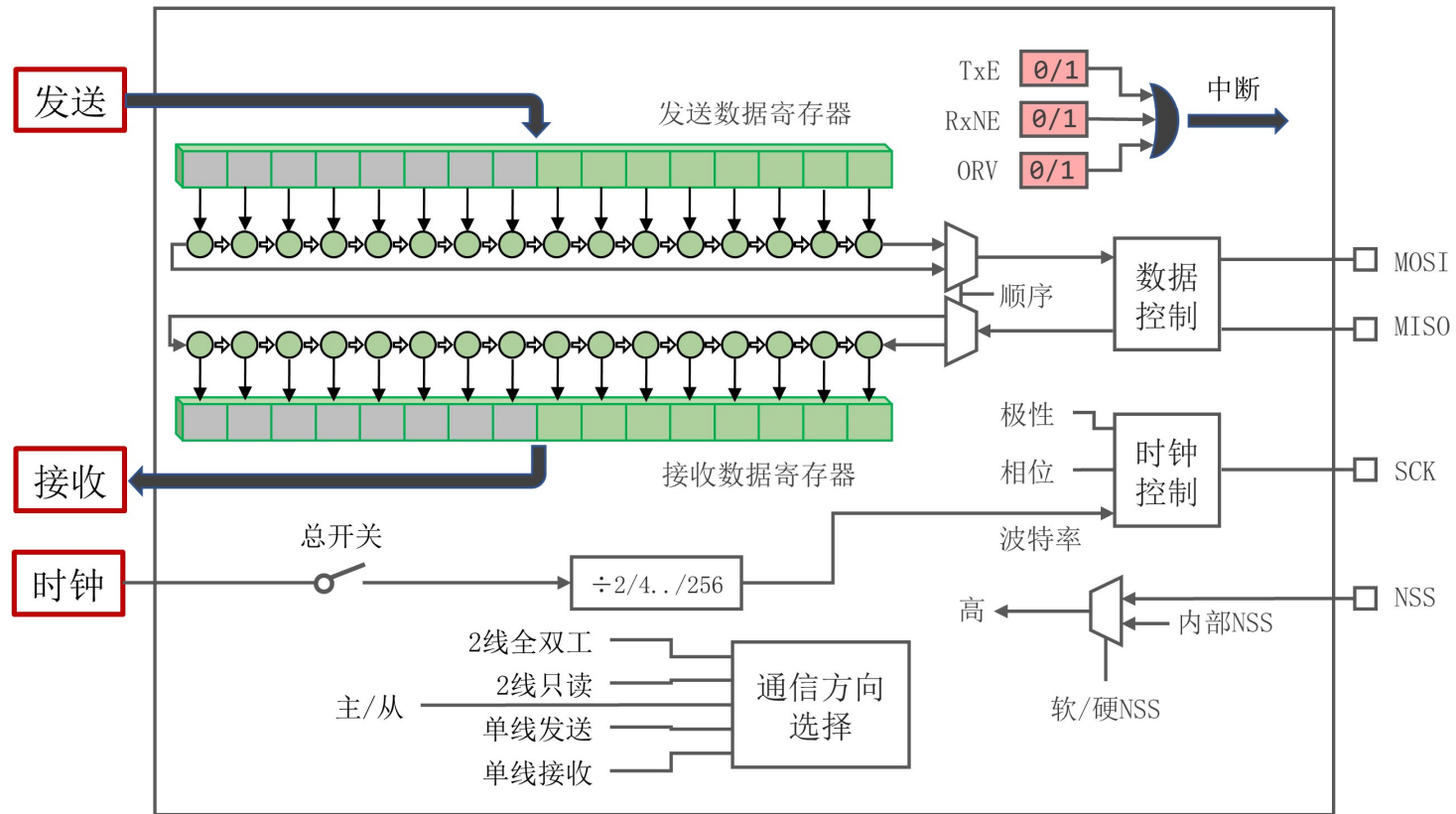
作从机时NSS作为片选输入
作主机时NSS应输入高电压



多主机模式
NSS被拉低时主机身份变为从机



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```
SPI_NSSInternalSoftwareConfig(SPI1, SPI_NSSInternalSoft_Set);
```

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